a. Shapes of bacteria

Based on the shapes, bacteria are grouped as:

1. Spherical shaped bacteria called as cocci (or coccus for a single cell).
2. Rod shaped bacteria called as bacilli (or bacillus for a single cell).
3. Spiral shaped bacteria called as spirilla (or spirillum for single cell).

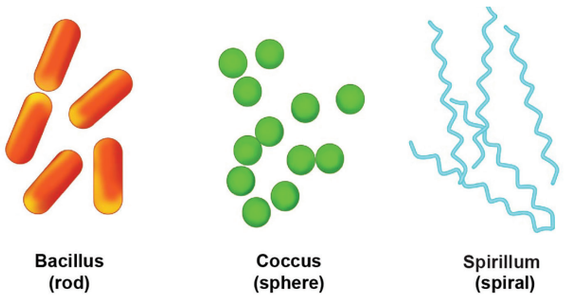


Figure 22.2 Shapes of bacteria

DO YOU KNOW?

Antonie Van Leeuwenhoek, the first microbiologist designed his own microscope. In 1674, he took plaque from his own teeth and observed it under the microscope. He was astonished to see many tiny organisms moving around, which was otherwise invisible to naked eyes.

b. Structure of a bacterial cell

Bacterial cell has cell membrane, covered by strong rigid cell wall. In some bacteria, outside the cell wall there is an additional slimy protective layer called **capsule** made up of **polysaccharides**. The plasma membrane encloses the cytoplasm, **incipient nucleus** (nucleoid), ribosomes and DNA which serve as genetic material. Ribosomes are the site of protein synthesis. They lack membrane bound organelles. In addition to this, a small extra chromosomal circular DNA called plasmid is found in the cytoplasm.

22.1.2 Viruses

The term 'virus' in Latin means 'venom' or 'poisonous fluid'. Viruses are non-cellular, **self-replicating parasites**. They are made up of a **protein** that covers a central **nucleic** acid molecule, either RNA or DNA. The amount of protein varies from 60% to 95% and the rest is nucleic acid. Nucleic acid is either DNA (T4 bacteriophage) or RNA (Tobacco mosaic virus, TMV).

A simple virus particle is often called a **virion**. They grow and multiply only in living cells. They are the smallest among the infective agents varying over a wide range from 18-400 nm (nanometre). They can live in plants, animals, human being and even bacteria. They can be easily transmitted from one host to another.

a. Characters of Viruses

Viruses exhibit both living and non-living characters.

Living characters of viruses

1. They have the nucleic acid (DNA or RNA) i.e., the genetic material that can replicate.
2. They can multiply in the living cells of the host.
3. They can attack specific hosts.

Non-living characters of viruses

1. Viruses remain as inert material outside their hosts.
2. They are devoid of cell membrane and cell wall. Viruses are devoid of cellular organelles like ribosomes, mitochondria, etc.
3. They can be crystallised.

More to Know

The protein free pathogenic RNA of virus is Viroids. They are found in plant cells and cause disease in plants.

b. Types of Viruses

Viruses are categorised as given below:

Plant virus: Virus that infect plants. e.g. Tobacco mosaic virus, Cauliflower mosaic virus, Potato virus.

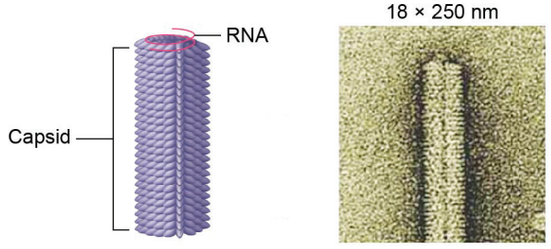


Figure 22.3 Tobacco mosaic virus

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